1. ***This is the start of a multipart question. The information below pertains to question 1 to question 8.***

   The data set `bwght` contains data on births in the US from the 1988 National Health Interview Survey. You can load the data into R using the `wooldridge` library using:

   ```
   library(wooldridge)
   dat = wooldridge::bwght
   ```

   The main outcome of interest is the variable *bwght*, which is the birth weight of the child in ounces.
   What is the mean of the variable *bwght* in the dataset?

   a. 120.0000
   b. 118.6996
   c. 116.9325
   d. 119.5121
   e. None of the above

2. How many missing values or NA's are there for the variable *fatheduc* in the dataset?

   a. 0
   b. 1
   c. 196
   d. 197
   e. None of the above

3. Consider the following linear regression model.

$$\begin{aligned} bwght = \beta_0 + \beta_1(cigs) + \beta_2(faminc) + \beta_3(faminc^2) + \beta_4(parity) + \\ \beta_5(motheduc) + \beta_6(fatheduc) + \beta_7(male) + \beta_8(white) + \varepsilon \end{aligned}$$

   The model includes the number of cigarettes smoked per day by the mother during pregnancy (*cigs*), family income and family income squared, and the birth order of the child (*parity*). The model also controls for the education of the mother and father. Finally, (*male, white*), indicator variables for the sex and race of the child are included in the model.
   Estimate the model above with OLS. What is the estimated coefficient on *cigs*?

   a. –0.4634
   b. –0.5950
   c. –0.5137
   d. –0.6211
   e. –0.4772

4. What is the sum of squared errors of the model?

   a. 461992.3

b. 13340.2
c. 390114.8
d. 511023.7
e. None of the above

5. Suppose we wanted to create a confidence interval for one of the parameters of the model. If we wanted to create a 99% confidence interval, what critical value would we use?

a. 2.5758
b. 2.3295
c. 2.5800
d. 1.9620
e. None of the above

6. Consider the null hypothesis that $H_0 : \beta_{male} = 0$. What is the strongest conclusion we can draw about this null hypothesis?

a. Reject the null at the 0.1% level
b. Reject the null at the 1% level
c. Reject the null at the 5% level
d. Reject the null at the 10% level
e. Fail to reject the null at the 10% level

7. Consider the null hypothesis that $H_0 : \beta_{cigs} = -0.5$. What is the p-value associated with this hypothesis test?

a. 0.1935
b. 0.3871
c. 0.6129
d. 0.0000
e. None of the above

8. What is the covariance between the estimate of the parameter on *(Intercept)* and the parameter on *white*?

a. –0.9241
b. 2.6122
c. –1.6775
d. 1.6775
e. –0.0017

9. ***This is the start of a multipart question. The information below pertains to question 9 to question 15.***

The data set `k401ksubs` contains data on workers from the 1991 Survey of Income and Program Participation. You can load the data into R using the `wooldridge` library using:

```
library(wooldridge)
dat = wooldridge::k401ksubs
```

The variable of interest is *e401k* which indicates if the worker is eligible for a 401(k) plan through their employer. We would like to study what are the main determinants of 401(k) eligibility using a linear probability model. To answer this question you will need to add a new variable to the data set called *young* which should equal 1 if the worker is younger than 35 (i.e., *age* < 35) and zero otherwise.
What percent of the population are *young*?

a. 28.42%
b. 68.31%
c. 35.11%
d. 31.69%
e. None of the above

10. Estimate a linear probability model on 401(k) eligibility *e401k*.
Include in your model controls for *young*, annual family income in $1000s (*inc*), and indicators for whether the worker is male (*male*) and married (*marr*). Your model should have 5 parameters including the intercept.
What is the R-squared from the estimated model?

a. 0.07355
b. 0.07394
c. 0.06112
d. 0.08227
e. 0.05930

11. What is the correct interpretation of the parameter on *young*? Holding fixed the variables in the model, being younger than 35 changes the probability of being eligible for a 401(k) by . . .

a. –4.0627 percent
b. –0.0406 percentage points
c. –4.0627 percentage points
d. –0.4063 percentage points
e. None of the above

12. What is the estimate of the upper bound of the 99% confidence interval for the parameter on *marr*?

a. –0.0015
b. –0.0543
c. 0.0092
d. 0.0057
e. None of the above

13. Consider the null hypothesis that $H_0 : \beta_{marr} = 0$. What is the strongest conclusion we can draw about this null hypothesis?

    a. Reject the null at the 0.1% level
    b. Reject the null at the 1% level
    c. Reject the null at the 5% level
    d. Reject the null at the 10% level
    e. Fail to reject the null at the 10% level

14. Re-estimate the model adding family size (*fsize*) as an additional control. What is the percentage change in the parameter on *marr* when we control for family size?

    a. –41.27%
    b. 41.27%
    c. –29.12%
    d. –70.27%
    e. None of the above

15. Re-estimate the model adding family size (*fsize*) as an additional control. What is the variance inflation factor for the parameter on *marr* when we control for family size? (note: if you are calculating the variance inflation factor from the formula you need to ensure that the same observations are used in the different regressions)

    a. 1.5284
    b. 1.1983
    c. 1.8029
    d. 0.5547
    e. None of the above

Solution Key

Q	Ans	Value	How / trap
1	**b**	118.6996	`mean(dat$bwght)`; 120 is the median
2	**c**	196	`sum(is.na(dat$fatheduc))`
3	**b**	–0.5950	lm drops 197 rows (196 *fatheduc* + 1 *motheduc*), n = 1191
4	**e**	456161.5	`sum(resid(m)^2)`; not listed
5	**c**	2.5800	`qt(0.995, df.residual(m))`, df = 1191 − 9 = 1182; 2.5758 is `qnorm`
6	**a**	$p = 0.00077$	$p < 0.001$
7	**b**	0.3871	$t = (\hat{\beta} + 0.5)/se = -0.866$; two-sided. 0.1935 is one-sided
8	**c**	–1.6775	`vcov(m)["(Intercept)","white"]`
9	**d**	31.69%	`young = as.numeric(age < 35)`; `<=35` gives 35.11%
10	**b**	0.07394	0.07355 is adjusted R^2
11	**c**	–0.0406	LPM: $\times 100$ = percentage points
12	**d**	0.0057	`confint(m1, level=0.99)`; 0.0092 is the 95% upper bound on *male*, –0.0015 the 95% upper on *marr*
13	**c**	$p = 0.0367$	$0.01 < p < 0.05$
14	**a**	–41.27%	100(–0.014287/ – 0.024325 – 1)
15	**c**	1.8029	`car::vif(m2)`; 0.5547 is $1 - R_j^2$

R code

```
# ECON 5021 Unit 2 Data Analysis Practice A: 26101200002 - solution key
# Key: 1b 2c 3b 4e 5c 6a 7b 8c 9d 10b 11c 12d 13c 14a 15c

library(wooldridge)
library(car)

# ---- Q1-8: bwght ----
dat = wooldridge::bwght

# Q1. What is the mean of the variable bwght in the dataset?
mean(dat$bwght)                                  # 118.6996 (b)

# Q2. How many missing values or NA's are there for the variable fatheduc?
sum(is.na(dat$fatheduc))                         # 196     (c)

# Q3. Estimate the model with OLS. Coefficient on cigs?
m = lm(bwght ~ cigs + faminc + I(faminc^2) + parity + motheduc + fatheduc + male + white, data =
     dat)
summary(m)                                       # -0.5950 (b)   n = 1191

# Q4. What is the sum of squared errors of the model?
```

```r
sum(resid(m)^2)                                  # 456161.5 (e)

# Q5. Critical value for a 99% confidence interval?
qt(1 - 0.01/2, df = df.residual(m))              # 2.5800  (c)   df = 1182, NOT 1388 - 9

# Q6. H0: b_male = 0. Strongest conclusion?
summary(m)$coefficients["male", ]                # p = 0.00077 -> reject at 0.1% (a)

# Q7. H0: b_cigs = -0.5. p-value?
t = (coef(m)["cigs"] - (-0.5)) / sqrt(vcov(m)["cigs", "cigs"])
2 * pt(-abs(t), df = df.residual(m))             # 0.3871  (b)

# Q8. Covariance between the estimates on (Intercept) and white?
vcov(m)["(Intercept)", "white"]                  # -1.6775 (c)

# ---- Q9-15: k401ksubs ----
dat = wooldridge::k401ksubs
dat$young = as.numeric(dat$age < 35)

# Q9. What percent of the population are young?
100 * mean(dat$young)                            # 31.69%  (d)

# Q10. LPM of e401k on young, inc, male, marr. R-squared?
m1 = lm(e401k ~ young + inc + male + marr, data = dat)
summary(m1)$r.squared                            # 0.07394 (b)

# Q11. Interpretation of the parameter on young?
coef(m1)["young"]                                # -0.040627 -> -4.0627 percentage points (c)

# Q12. Upper bound of the 99% CI for the parameter on marr?
confint(m1, level = 0.99)                        # 0.005662 (d)

# Q13. H0: b_marr = 0. Strongest conclusion?
summary(m1)$coefficients["marr", ]               # p = 0.0367 -> reject at 5% (c)

# Q14. Add fsize. Percentage change in the parameter on marr?
m2 = lm(e401k ~ young + inc + male + marr + fsize, data = dat)
100 * (coef(m2)["marr"] / coef(m1)["marr"] - 1)  # -41.27% (a)

# Q15. VIF for marr when we control for fsize?
vif(m2)["marr"]                                  # 1.8029  (c)
```